

Scaling Smart or Scaling Stupid?

A data-driven audit of CLV integrity, profitability leakage,
and risk-adjusted growth strategy across 1 million customers.

01 BIGGEST STRATEGIC RISK

CLV is overstated by \$17.3B.
Refunds are being counted as value,
not deducted from it.

02 BIGGEST HIDDEN OPPORTUNITY

23% of customers are loyal,
low-spend, and under-monetised.
Subscription is the unlock.

03 THREE EXECUTIVE DECISIONS

Fix CLV definition. Restructure
discounts by segment. Income-stratify
credit limits immediately.

1M

Customers Analysed

3

Data Sources Merged

\$17.3B

CLV Overstated

r=0.99

Refund ↔ CLV Gap

Revenue Is Real.

CLV Is Not.

ShopSphere's reported Customer Lifetime Value metric is structurally inflated — driven by refunds being counted as transactions rather than deducted as costs. Every decision built on this number is wrong.

\$17,277

Avg CLV Overstatement
per Customer

× 1M customers = \$17.3B phantom value

$r = 0.99$

CLV Gap explained
by Refund Amount
Near-perfect correlation
not a coincidence

$r = -0.66$

Return Rate vs.
Net Revenue/Month
High-revenue customers return the most

$r = -0.54$

Return Rate vs.
Adjusted CLV
Real value destroyed by return behaviour

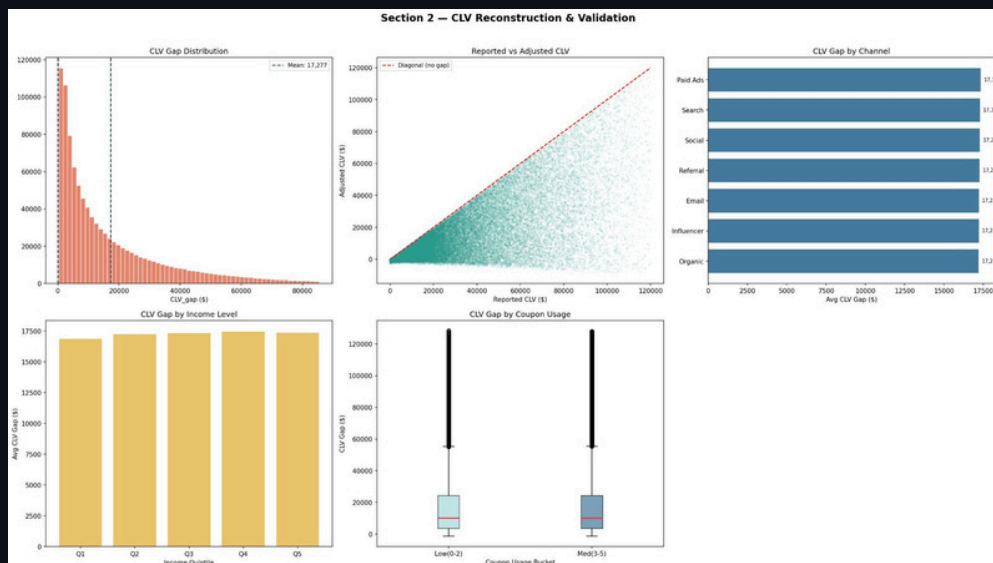


Fig 1 — CLV Gap distribution (left), Reported vs Adjusted CLV scatter (centre), and CLV Gap by acquisition channel (right)

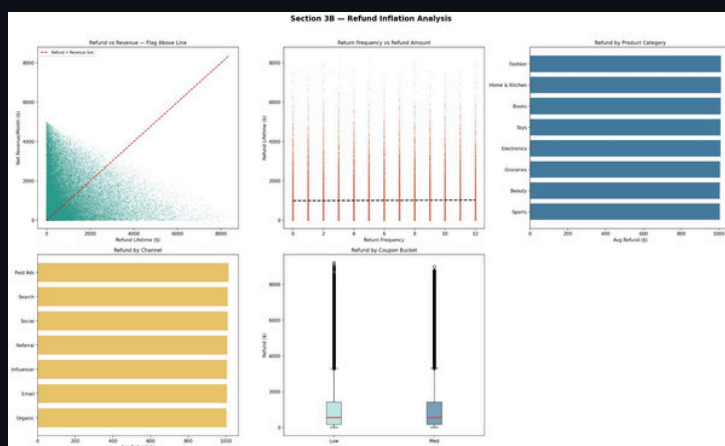


Fig 2 — Refund vs Revenue scatter (customers above the diagonal are net-negative), return frequency vs refund amount, and refund by channel

BOARD-LEVEL RISK STATEMENT

ShopSphere's CLV framework does not deduct refund_amount_lifetime. The CLV gap correlates with refunds at $r=0.99$. Every channel ROI model, retention budget, and performance target built on reported CLV is measuring phantom profit.

The Loyal Low-Spend Segment Is Your Real Growth Engine.

23% of ShopSphere's customers—230,000 people—have high loyalty, low discount dependency, and subscription adoption, but are generating below-median revenue. They are not low-value. They are under-activated.

27%

Champions

High CLV
High Loyalty
Defend & upsell

23%

Loyal Low-Spend

Hidden Gold
Subscription-ready
Activate now

23%

High-Value At-Risk

High CLV
High Churn Risk
Frail foundation

27%

Disengaged

Low CLV
Low Loyalty
Minimise cost

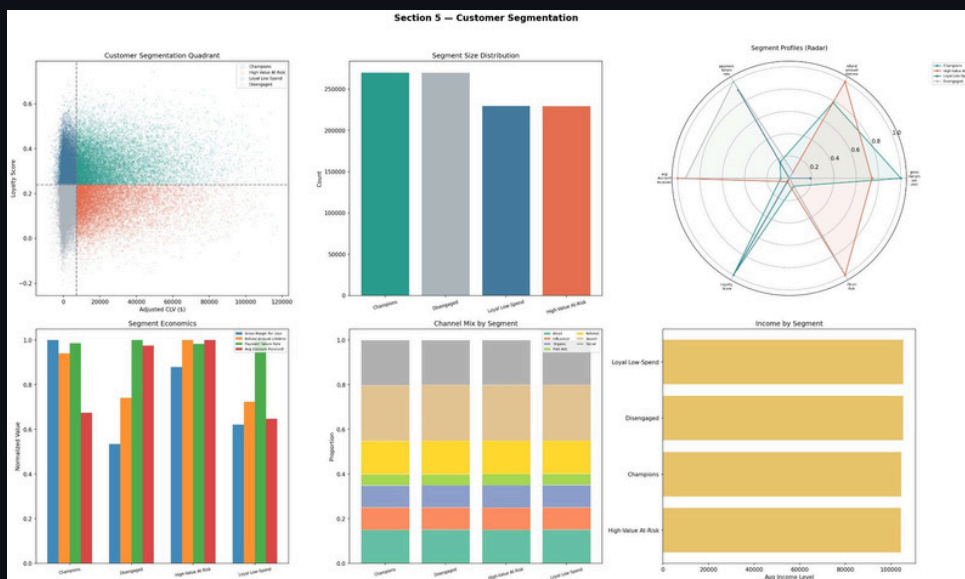


Fig 3 — Customer segmentation quadrant (Adjusted CLV × Loyalty Score), segment economics, channel mix, and radar profiles

Why Subscription Is the Unlock

Subscription holders show 20% higher Loyalty Scores and materially lower Churn Risk scores ($r = -0.36$ subscription vs churn). Premium subscribers have higher Adjusted CLV and stronger wallet share percentage. The Loyal Low-Spend cohort has the behavioural profile of a subscriber — consistent purchase cadence, brand loyalty score above median, low promo sensitivity — but has not yet been converted.

The psychographic data reveals a critical ethics flag: stress_from_financial_decisions correlates with impulse_purchases_per_month at $r = 0.74$. If marketing campaigns are optimised toward promo-sensitive, financially stressed customers, ShopSphere may be extracting short-term gains from its most vulnerable segment. The board must decide: is this a growth strategy or a liability?

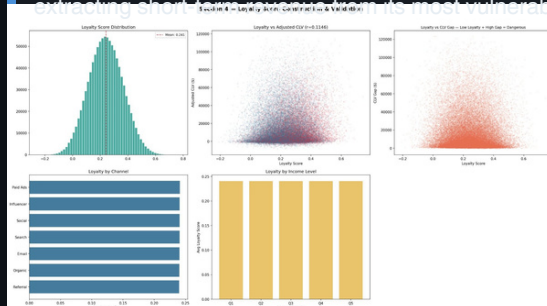


Fig 4 — Loyalty Score distribution, Loyalty vs Adjusted CLV scatter, and loyalty by channel. Referral channel produces lowest loyalty — the attribution paradox.

Stop Measuring Gross. Start Managing Net.

Threedecisionsthatcanbeoperationalisedwithin30days,eachtargetinga confirmed data-backed leakage point.

01 Restate CLV Net of Refunds and Risk Costs

WHY NOW

Every performance metric, channel ROI calculation, and retention budget is currently built on a CLV definition that does not subtract refund_amount_lifetime, payment failure exposure, or chargeback costs. The CLV gap has a near-perfect $r=0.99$ correlation with refunds alone. This is not a modelling nuance — it is a foundational accounting error that flows into every downstream decision.

ACTION

Adjusted CLV = (net_revenue_per_month × active_months) – avg_discount_received – refund_amount_lifetime – (payment_failure_rate × credit_limit) – (chargeback

IMPACT: Reranks top-performing channels and segments. Eliminates phantom profitability from the P&L narrative.

02 Restructure Discounts by Segment — Not by Calendar

WHY NOW

Gross margin drops monotonically from ~\$850 (lowest discount quartile) to ~\$510 (highest discount quartile) — a 40% margin compression. Critically, even the highest brand_loyalty_score quartile receives equivalent avg_discount_received as the lowest loyalty quartile. ShopSphere is donating margin to customers who would purchase without a discount. Meanwhile, High-Value At-Risk customers have the highest coupon_usage_frequency of any segment — discounts are deepening dependency, not loyalty.

ACTION

Champions & Loyal Low-Spend → Remove blanket discounts. Offer experiential value (early access, priority service).
High-Value At-Risk → Time-bounded, category-specific offers with return restrictions.
Disengaged → No discount investment. Let them exit or convert organically.

IMPACT: Estimated margin recovery: \$340 per customer moved from high to low discount dependency tier.

03 Income-Stratify Credit Limits and Payment Risk Exposure

WHY NOW

payment_failure_rate correlates with income_level at $r=-0.49$. Income Q1 customers carry median payment failure rates nearly double those of Q5 customers. Current credit_limit distribution appears uniform across income segments. This means ShopSphere is extending the same credit rope to customers with materially different ability to repay — concentrating chargeback and bad debt risk in the lowest-income cohort, while losing the loyalty-building opportunity with high-income customers who could absorb larger limits productively.

ACTION

Q1-Q2 income: Reduce credit_limit by 30-40%. Add payment plan friction. Flag for risk review.
Q3 income: Maintain current limits. Monitor payment_failure_rate quarterly.
Q4-Q5 income: Increase credit_limit selectively. Target for subscription conversion.

IMPACT: Reduces expected credit loss. Frees capital for Loyal Low-Spend subscription activation campaigns.

Supporting Evidence

Key correlations, master data map, and analytical validation underpinning all board-level findings.

Top Confirmed Signal Relationships

Relationship	r Value	Strength	Strategic Implication
CLV Gap ↔ refund_amount_lifetime	+0.99	CRITICAL	CLV is a refund proxy, not a value metric
customer_lifetime_value ↔ refund_amount_lifetime	+0.74	STRONG	Reported CLV rises with refunds — backwards
net_revenue_per_month ↔ return_rate	-0.66	STRONG	Revenue leaders are return leaders too
Adjusted_CLV ↔ net_revenue_per_month	+0.78	STRONG	Adjusted CLV is grounded in real cash flow
Adjusted_CLV ↔ return_rate	-0.54	MODERATE	Return behaviour destroys real value
income_level ↔ payment_failure_rate	-0.49	MODERATE	Lower income = higher credit risk
Loyalty_Score ↔ wallet_share_percentage	-0.43	MODERATE	Loyal customers concentrate spend here
Loyalty_Score ↔ Churn_Risk	+0.58	MODERATE	Loyalty score is a valid churn predictor
avg_discount_received ↔ gross_margin_per_user	-0.25	WEAK-MOD	Discounts measurably erode margin
stress_financial ↔ impulse_purchases_per_month	+0.74	STRONG	Ethics risk: stressed users buy impulsively
credit_limit ↔ payment_failure_rate	+0.74	WEAK-MOD	Uniform limits hide income-based risk
subscription_revenue_share ↔ Loyalty_Score	-0.39	WEAK	Subscription adoption signals loyalty
	+0.31		

Master Correlation Matrix — 18 Priority Variables (Analytical North Star)

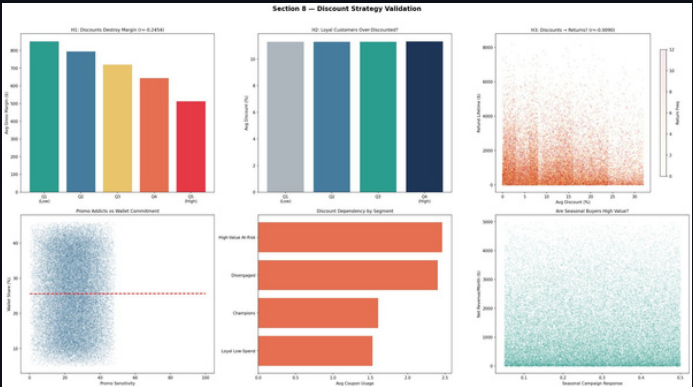
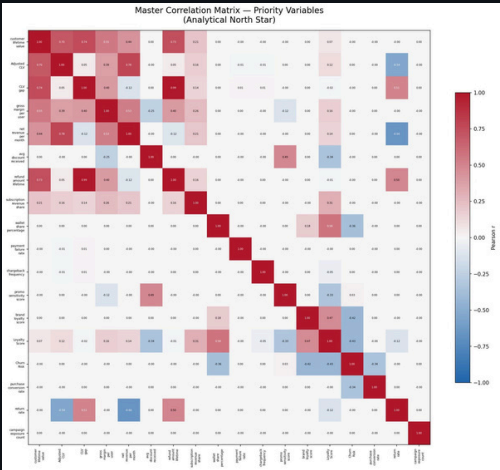


Fig 5 — Discount validation (H1/H2/H3 hypothesis charts)

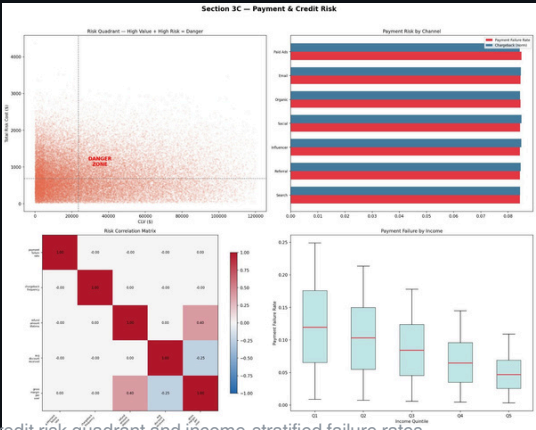


Fig 6 — Payment & credit risk quadrant and income-stratified failure rates